

selling volatility protection (again, e.g., in the form of selling put options). Buying or selling volatility protection can be done in option markets, but traders can also use other derivatives contracts, such as volatility swaps. Investors are so concerned about volatility, on average, that they are willing to pay to avoid volatility risk, rather than be paid to take it on. Periods of high volatility coincide with large downward movements (see Figure 7.3) and assets that pay off during high volatility periods, like out-of-the-money puts, provide hedges against volatility risk.

We often think about assets having *positive* premiums—we buy, or go long, equities, and the long position produces a positive expected return over time. Volatility is a factor with a *negative* price of risk. To collect a volatility premium requires *selling* volatility protection, especially selling out-of-the-money put options. The VIX index trades, on average, above volatilities observed in actual stocks: VIX implied volatilities are approximately 2% to 3%, on average, higher than realized volatilities. Options are thus expensive, on average, and investors can collect the volatility premium by short volatility strategies. Fixed income, currency, and commodity markets, like the aggregate equity market, have a negative price of volatility risk.<sup>11</sup>

Selling volatility is not a free lunch, however. It produces high and steady payoffs during stable times. Then, once every decade or so, there is a huge crash where sellers of volatility experience large, negative payoffs. Figure 7.4 plots the cumulated returns of a volatility premium (swap) index constructed by Merrill Lynch. There are steady returns until 2007, with the few blips corresponding to some small losses during 1998 (Russian default crisis) and 2001 and 2002 (9/11 tragedy and economic recession, respectively), and also during the summer of 2007 (subprime mortgage-backed losses just prior to the financial crisis). But between September and November 2008 there are massive losses close to 70%. These were the darkest months in the financial crisis, and most of the losses across all types of risky assets during 2008 were concentrated during these months (see Table 6.1). The huge crash causes volatility selling to have a large negative skewness of  $-8.26$  over the whole sample. Taking data prior to the financial crisis ending December 2007, the skewness is only  $-0.37$ , so prior to 2007 selling volatility looked like easy money.

<sup>11</sup> For a negative volatility risk premium in fixed income markets, see Simon (2010) and Mueller, Vedolin, and Yen (2012); currency markets, see Low and Zhang (2005); commodity markets, see Prokopczuk and Wese (2012); and the aggregate stock market, see Bakshi and Kapadia (2003) and Ang et al. (2006). For individual stocks, the volatility risk premium can be positive (some agents really like individual stock risk), see Driessen, Maenhout, and Vilkov (2009). One explanation why individual stocks can carry positive risk premiums but the volatility risk premium is significantly negative at the aggregate level is that much of a stock's variation is idiosyncratic (see also chapter 10). In portfolios, the stock-specific, idiosyncratic movements are diversified away leaving only market volatility risk, which has a negative risk premium. In the discussion in the main text, volatility risk refers to both "smooth" movements in time-varying volatility (*diffusive* risk) and abrupt changes (*jump* risk). More sophisticated models differentiate between the two; see Pan (2002).